

Accessibility Tools Findings

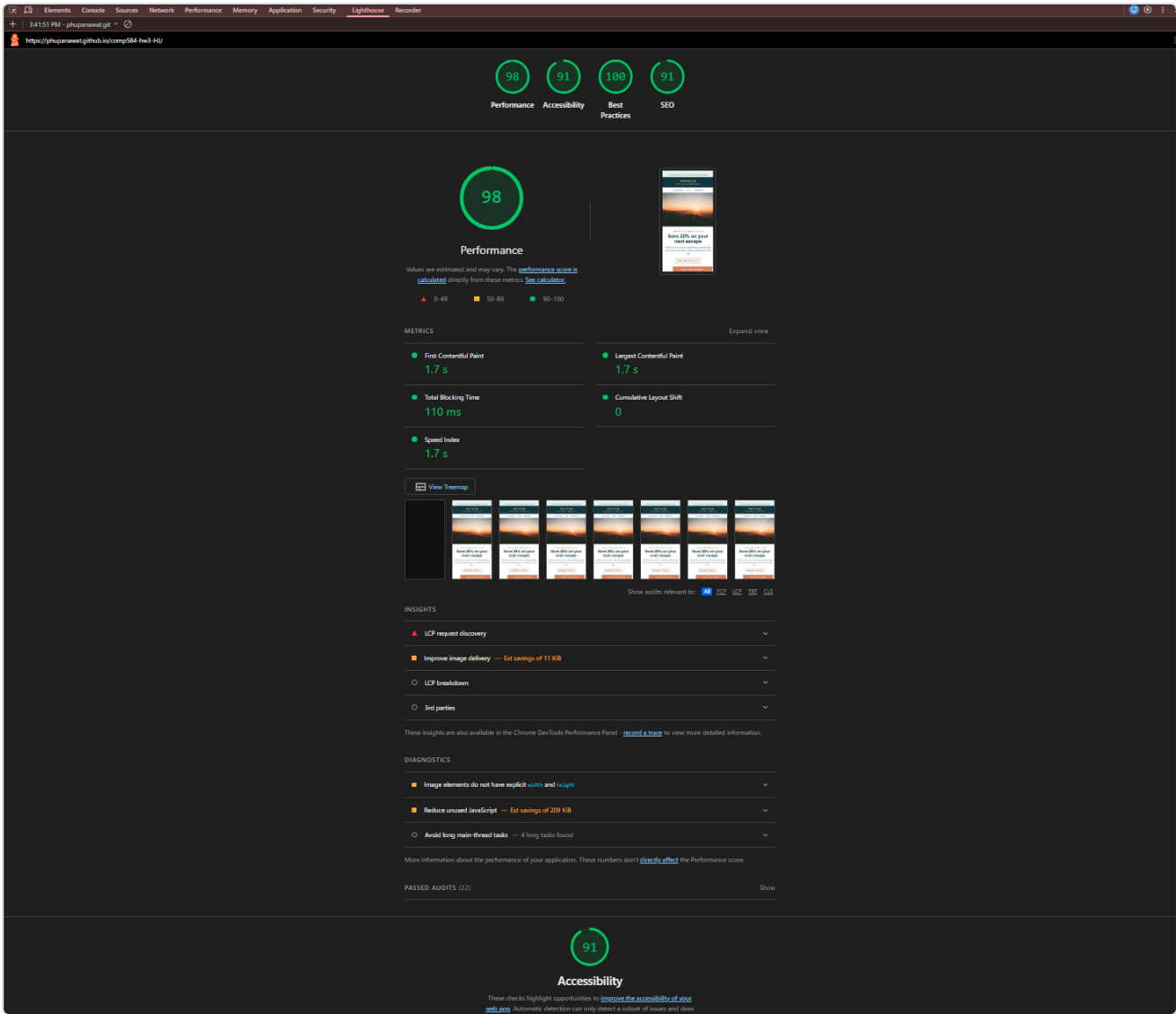
COMP 584 - HW3 · Tools explored: Chrome DevTools (Lighthouse), Axe DevTools, Silktide

Site tested	Lighthouse (Accessibility)	Axe	Silktide
My HW3 email (this assignment)	91	5 issues (color contrast)	15 issues (4 categories)
youtube.com	100	0 issues	Failed to scan - crashed
twitch.tv	100	Not tested	Failed to scan - crashed

1. My HW3 Email

Chrome DevTools — Lighthouse

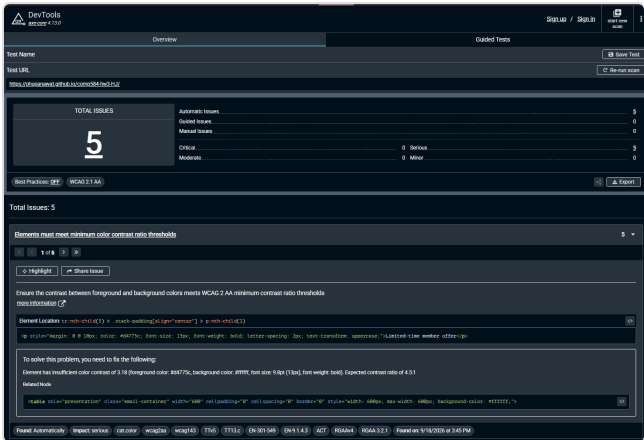
Target: <https://phupanawat.github.io/comp584-hw3-HJ/>



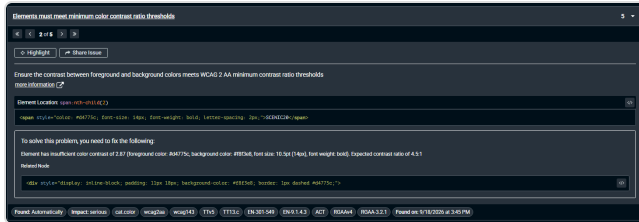
Lighthouse report - Accessibility score 91/100

Axe DevTools

5 automatic issues found, all the same rule: "Elements must meet minimum color contrast ratio thresholds" (serious, WCAG 2 AA 1.4.3)



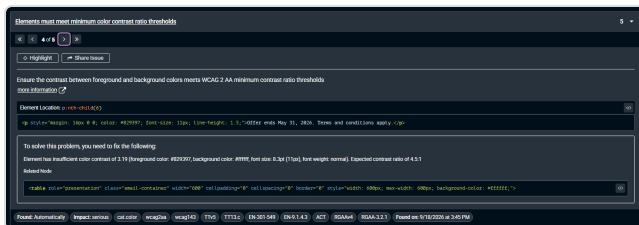
Issue 1/5 - "Limited-time member offer" label, contrast 3.18 (needs 4.5:1)



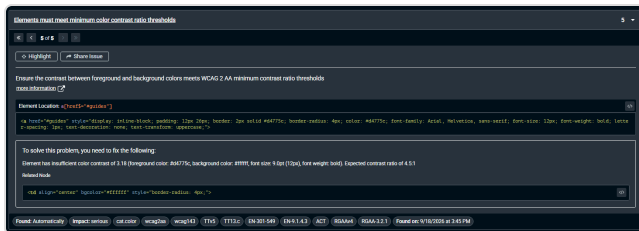
Issue 2/5 - "SCENIC20" promo code, contrast 2.87



Issue 3/5 - "Book your getaway" button text, contrast 3.18



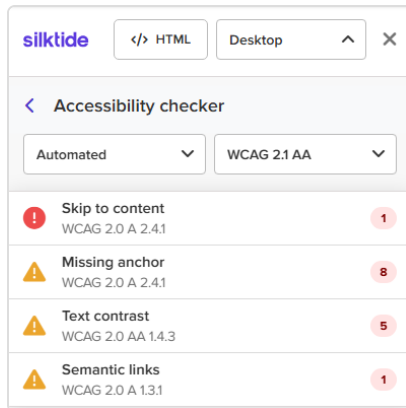
Issue 4/5 - "Offer ends..." fine print, contrast 3.19



Issue 5/5 - "Read the travel guide" button text, contrast 3.18

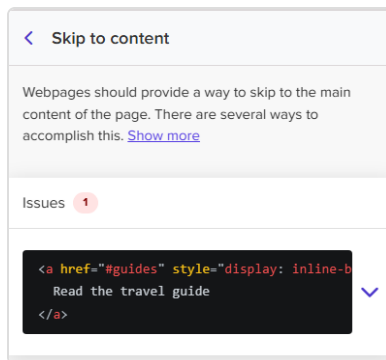
Silktime

15 issues across 4 categories



Accessibility checker	
Automated	WCAG 2.1 AA
 Skip to content WCAG 2.0 A 2.4.1	1
 Missing anchor WCAG 2.0 A 2.4.1	8
 Text contrast WCAG 2.0 AA 1.4.3	5
 Semantic links WCAG 2.0 A 1.3.1	1

Overview: Skip to content (1), Missing anchor (8), Text contrast (5), Semantic links (1)



< Skip to content

Webpages should provide a way to skip to the main content of the page. There are several ways to accomplish this. [Show more](#)

Issues **1**

```
<a href="#guides" style="display: inline-b...  
Read the travel guide  
</a>
```

"Skip to content" - flags no skip-navigation link

<

Missing anchor

Do not link to an anchor on the page that does not exist. For example: linking to #content without a valid destination such as <body id="content"></body>.

Show more

Issues

8

<a href="#browser" style="color: #557078;

<a href="#destinations" style="margin: 0 1

Destinations

<a href="#stays" style="margin: 0 12px; co

Stays

<a href="#inspiration" style="margin: 0 12

Inspiration

<a href="#explore" class="mobile-button" s

Book your getaway

<a href="#guides" style="display: inline-b

Read the travel guide

<a href="#preferences" style="color: #f5c7

<a href="#unsubscribe" style="color: #f5c7

"Missing anchor" - 8 links point to in-page IDs that don't exist (placeholder hrefs)

<

Text contrast

To comply with WCAG AA, the colour of text must sufficiently contrast with its background color, so that people with moderate visual impairments can read it.

Show more

Issues

5

Limited-time member offer

SCENIC20

Book your getaway

Offer ends May 31, 2026. Terms and con...

Read the travel guide

"Text contrast" - same 5 low-contrast elements Axe found

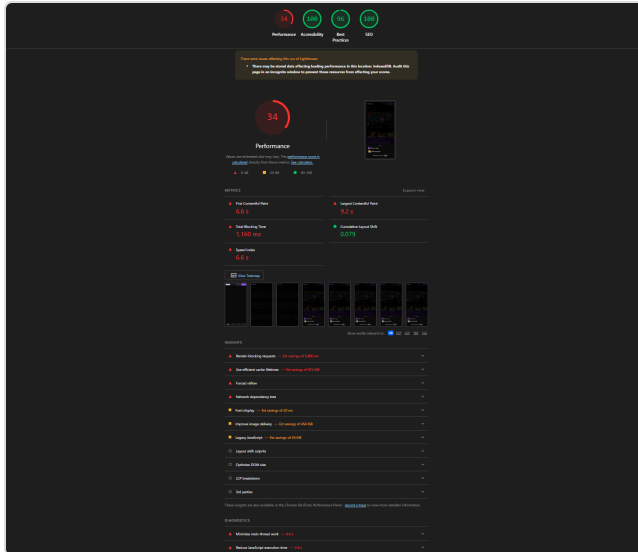


"Semantic links" - nav links should be marked up as a list

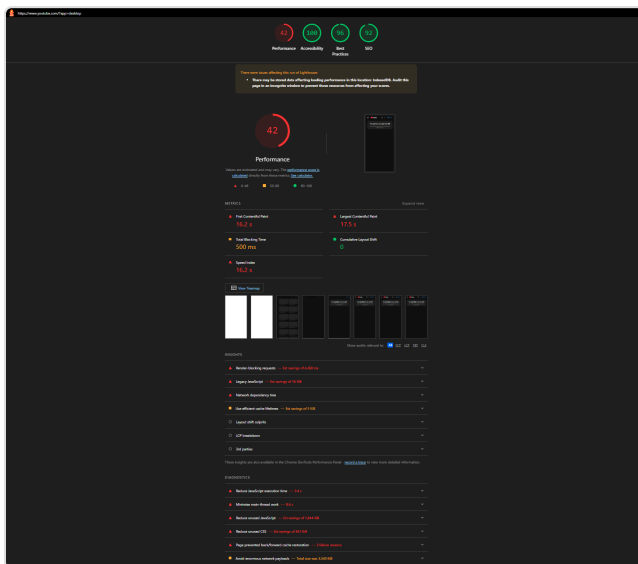
Takeaway: Two independent tools (Axe and Silktide) agreed on the same root cause. My accent color #d4775c doesn't reach 4.5:1 contrast against white/cream backgrounds. That's a genuine, fixable bug and the clearest signal from this whole exercise: when multiple tools flag the identical issue, it's real, not a false positive.

2. Sites I Use Regularly

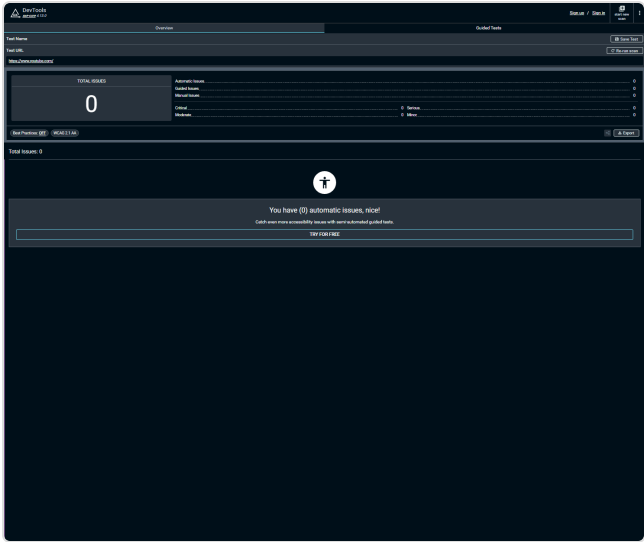
YouTube (youtube.com)



Lighthouse - Accessibility 100, Performance 34



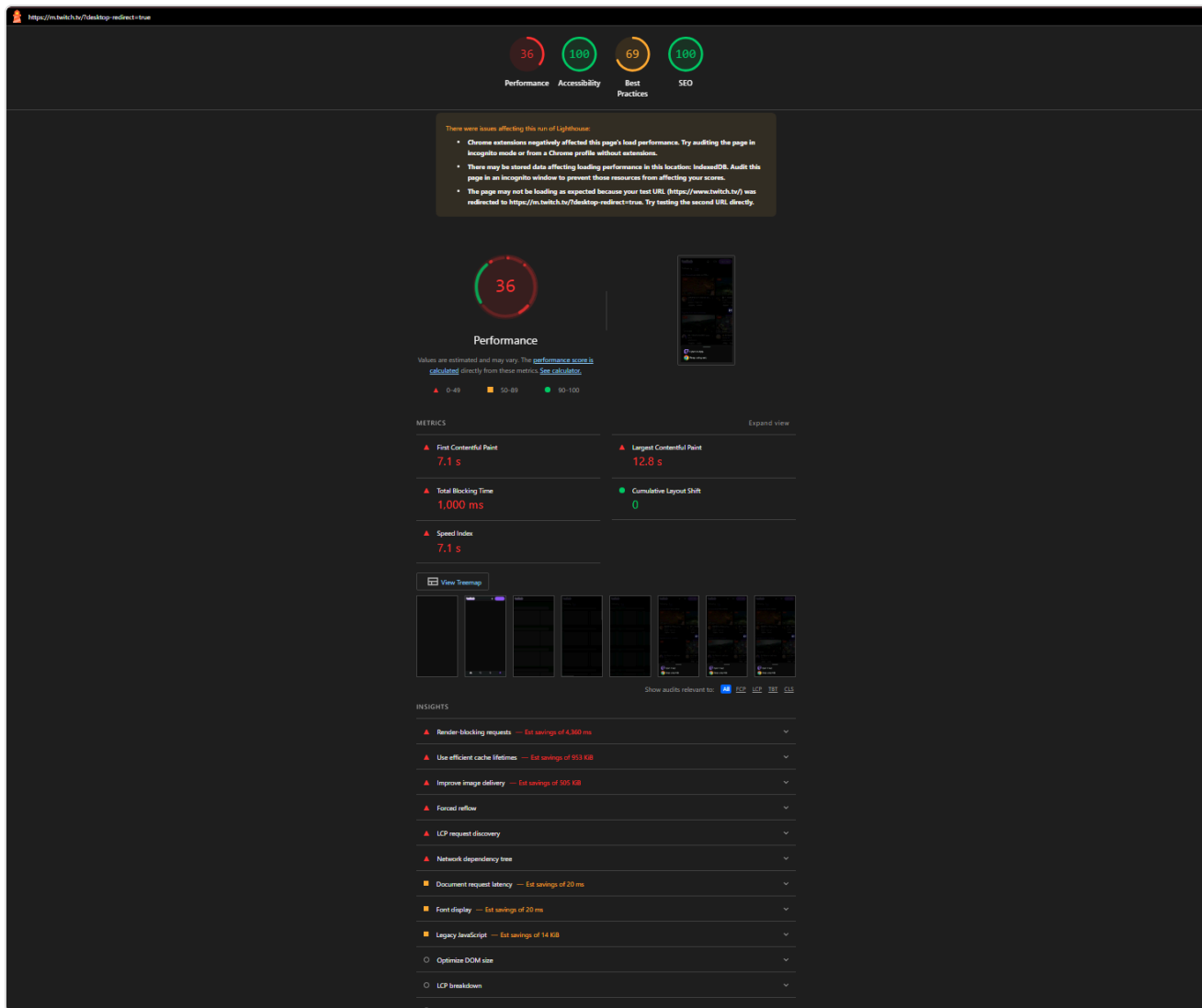
Lighthouse re-run - Accessibility 100, Performance 42



Axe DevTools - 0 automatic issues found

Silktide would not load/scan this page - the extension repeatedly crashed on youtube.com.

Twitch (twitch.tv)



Lighthouse - Accessibility 100, Best Practices 69, Performance 36

Silktide would not load/scan this page either - same crash behavior as on YouTube.

3. Pros & Cons by Tool

CHROME DEVTOOLS (LIGHTHOUSE) — PROS

- Built into Chrome, no extension or account needed
- Scores accessibility alongside Performance/Best Practices/SEO, so you see it in context of overall page health
- Fast, one-click "Analyze page load"

CHROME DEVTOOLS (LIGHTHOUSE) — CONS

- Shallow automated checks - gave both YouTube and Twitch a perfect 100 despite likely real issues on such large, complex sites
- No live overlay on the page; you don't see issues in context without switching panels

- Doesn't point at the exact failing element the way Axe does

AXE DEVTOOLS — PROS

- Cites the exact WCAG success criterion (e.g. 1.4.3) and computes the precise contrast ratio vs. the ratio required
- Shows the actual violating HTML snippet inline, so fixes are unambiguous
- Handled every site I tested, including JS-heavy ones like YouTube, without issue

AXE DEVTOOLS — CONS

- Only reports "automatic" issues by default; deeper "Guided Tests" (keyboard nav, screen reader flow) are gated behind sign-up
- Dense, developer-facing UI — less skimmable at a glance than Silktide's

SILKTIDE — PROS

- Plain-language explanations of each issue (good for non-technical stakeholders)
- Simple overlay grouped by issue type with counts — easy to scan at a glance
- Caught the same contrast problems Axe did, confirming the finding

SILKTIDE — CONS

- **Reliability:** crashed/failed to scan both YouTube and Twitch entirely — couldn't handle JS-heavy, large sites that Axe and Lighthouse scanned without trouble
- Flags generic full-website conventions (like "skip to content") that don't map cleanly onto a single-page context like an HTML email, where that pattern isn't standard practice or well-supported by email clients